Truly Subcubic Exact Triangle over Bounded-Exponent Abelian Groups

ZIJIAN ZENG, UCSI University, Malaysia

For every fixed integer $M \geq 2$, we give a deterministic algorithm for Exact Triangle over compactly presented finite abelian groups of exponent dividing M and polynomial order. On a complete tripartite graph with at most n vertices in each part and group order at most n^c, the algorithm runs in $O_{M,c}(n^{3-\delta_M} \operatorname{polylog}(n+2))$ word-RAM operations, where $\delta_M > 0$ is independent of c. This resolves the bounded-exponent question in Majdoddin's Open Problem 21.12, using his elementary-group algorithm as the base case. One conservative saving is $\delta_M = 10^{-3}(144000)^{1-\Omega(M)}$, where Ω counts prime factors with multiplicity. We construct quotient-zero pieces using capped witness lists and the known hitting-set/frequency decomposition. Each piece has edgewise vertex potentials. Lifting these potentials cancels carries before division into the kernel, even for non-split extensions. Separate absence markers make every recursive query a legitimate complete input. All witness acquisition, graph construction, coordinate arithmetic and recursive calls are charged.

Additional Key Words and Phrases: Exact Triangle, finite abelian groups, bounded exponent, fine-grained algorithms, quotient and kernel

Author's Contact Information: Zijian Zeng, UCSI University, Malaysia, 1002266693@ucsiuniversity.edu.my.

1 The result and its context

Exact Triangle asks whether three edge labels around a triangle sum to zero. The algebra of the labels matters when the group is given by short coordinate vectors rather than by an enumerated multiplication table. We consider a complete tripartite graph with parts U, X, V, each of size at most n. Every edge has a label in

$$G = \prod_{j=1}^{d} \mathbb{Z}_{m_j}, \qquad m_j \geq 2, \quad m_j \mid M, \qquad |G| \leq n^c.$$

The presentation lists the cyclic factors and each label's residues. Both $M \geq 2$ and $c > 0$ are fixed, but the algorithm must handle every dimension allowed by this order bound. A solution is a triple whose three labels sum to zero in G. We allow the empty product, representing the trivial group.

Theorem 1.1. *For each fixed $M \geq 2$ there is a deterministic uniform word-RAM algorithm that solves this problem, for every fixed $c > 0$, in*

$$O_{M,c}\big(n^{3-\delta_M} \operatorname{polylog}(n+2)\big), \qquad \delta_M = \frac{1}{1000}\left(\frac{1}{144000}\right)^{\Omega(M)-1}.$$

The saving is independent of c. The bound includes reading the input, preprocessing, constructing every instance, coordinate arithmetic, and all recursive calls. Words have $\Theta(\log(n+2))$ bits. Each group vector has $O_c(\log(n+2))$ residues, and operations are charged coordinate by coordinate.

Majdoddin [2], Theorem 21.3 and Corollary 21.7(a), supplies the elementary-group case. His Remark 21.8 explains the difficulty in transferring the list-based lifting method to triangles: an edge's lift carry depends on the other labels, and a useful split must survive re-gauging. His Propositions 21.9 and 21.10 treat, respectively, few quotient-zero triangles and limited incident residue diversity. Open Problem 21.12 asks for a truly subcubic algorithm without these promises, including the unrestricted $\mathbb{Z}_4^d$ case and the general fixed bounded-exponent case. Theorem 1.1 extends the elementary backend to every fixed bounded exponent for compact cyclic-product inputs.

Our contribution is the fully charged quotient/kernel extension step. Chan, Vassilevska Williams and Xu [1, Theorem 5.1] give a zero-triangle decomposition with a hitting-set/frequency structure. Its initial witness sorting is cubic; its faster update bound cannot be used as fresh-input preprocessing. We reconstruct the part of that structure needed here using capped detection, and explicitly expose the edgewise potential equations already underlying their construction. Those equations let us cancel arbitrary lift carries before making kernel queries. We pay for complete, explicitly labelled graphs rather than using the cost of a compact representation. A finite induction then handles every fixed M. Capped listing, hitting sets and frequency decomposition are established supporting tools. For prior relationships between detection, all-edge detection and listing, see Sheffield, Vassilevska Williams and Xi [3]. Our extension uses the conservative capped routine proved below.

The result removes the elementary/non-elementary distinction for this entire bounded-exponent class, including dense quotient-zero inputs with many incident residue classes. The quantitative guarantee is weak: $\delta_4 = 1/144000000$, and it deteriorates with $\Omega(M)$. The comparison above uses the stated versioned primary results.

Research methodology. OpenAI Codex materially contributed to developing the quotient/kernel algorithm and proof, comparing prior work, and implementing exact semantic checks. A separate model context

reviewed the mathematical argument and wrote an additional verifier. The closing AI methodology statement describes the full scope of this assistance.

2 Representation and the elementary backend

Throughout, $\widetilde{O}$ suppresses a fixed power of $\log(n+2)$ and constants depending on the fixed parameters. Group coordinates have bounded modulus when M is fixed. The order promise gives $d \le c\log_2 n$ for nontrivial inputs. Thus coordinatewise operations and comparisons cost $O_{M,c}(\log(n+2))$, not one operation. Sorting means deterministic lexicographic sorting of representations, without assuming an order compatible with group addition.

The embedding $\mathbb{Z}_m \hookrightarrow \mathbb{Z}_M$, $x \mapsto (M/m)x$, is injective when $m \mid M$. Applying it to every coordinate preserves zero sums and takes $O_{M,c}(n^2\log(n+2))$ operations. The ambient group becomes $\mathbb{Z}_M^d$, of order at most $n^{c\log_2 M}$. There is no group enumeration. We may therefore prove the algorithm for homogeneous groups.

We use the following precise prior dependency, with $\varepsilon_0 = 1/1000$. For every fixed prime p, elementary-group Exact Triangle over $\mathbb{F}_p^D$ has a deterministic algorithm taking

$$O_{p,C}\big(b^{3-\varepsilon_0}\operatorname{polylog}(b+2)\big) \quad \text{whenever } p^D \le b^C, \tag{1}$$

for every fixed C, with the saving independent of C. Indeed, Majdoddin [2, Theorem 21.3] states the running time

$$O\big(p^6 b^{3-\varepsilon\gamma}(\log b)^6\big)$$

under his matrix primitive $H(\varepsilon,\gamma)$. His Corollary 9.1(b) and Section 16 instantiate $H(1/14, 0.077)$, giving $\varepsilon\gamma = 0.0055 > \varepsilon_0$; Corollary 21.7(a) makes this unconditional. For a fixed p, the restriction $p \le b^{\varepsilon/4}$ excludes only a finite range, handled by direct enumeration. We rely on this stated prior theorem and its matrix dependency, rather than assuming a new matrix oracle. The algorithm uses the actual dimension, not a supplied numerical bound C.

For the rest of the proof, write the labels on U_iX_k, X_kV_j, U_iV_j as $A_{ik}, B_{kj}, -C_{ij}$, respectively. We seek

$$A_{ik} + B_{kj} - C_{ij} = 0 \quad \text{in } \mathbb{Z}_M^d. \tag{2}$$

Missing edges arise only inside reductions; the original input is complete. Here is a uniform way to represent them. Over any $\mathbb{Z}_q^D$ with $q \ge 2$, append three coordinates, one for each edge kind. Present edges have zero markers. An absent edge has zero original coordinates and marker 1 in its own kind's coordinate, zero in the other two. These are the actual *plus-sum* labels. A triangle has one edge of each kind, so its marker sum is zero exactly when all three edges are present. The construction works also in characteristic two, where one shared sentinel could cancel. Encoding a graph of side b costs $O(b^2(D+3))$ operations.

3 Capped witnesses from legal detection queries

The following standard detection/listing construction records the cutoff needed when dimension is measured against the original input size.

Lemma 3.1. *Fix a prime p, constants $c > 0$ and $0 < \theta < 1$, and a positive integer cap $T = O(n^{1-\theta})$. On a possibly masked instance over $\mathbb{F}_p^d$ with $p^d \le n^c$, one can deterministically obtain, for each outer pair (i,j),*

exactly $\min(t_{ij}, T)$ *distinct middle witnesses, where* t_{ij} *is its true number of present zero triangles. The fully charged time is* $\widetilde{O}(n^{3-\varepsilon_0\theta/6})$.

Proof. Handle $n \le 1$ directly. Pad the three parts to the least power of two $N \ge n$, marking every edge incident to a padded vertex absent. Put $\alpha = \theta/6$, and let b_0 be the least power of two at least N^α. Thus $N^\alpha \le b_0 < 2N^\alpha$, except for the harmless equality at $N = 1$. Store a counter and a list for each original outer pair.

Traverse the product octree of the three padded intervals in depth-first order: each cube of side $b > b_0$ has eight children of side $b/2$. On entering a cube, construct a fresh complete encoded query. Its active edges are the original present edges, except that any outer pair whose counter has reached T is also absent. Query (1), using the three absence coordinates. A negative cube returns. A positive cube of side larger than b_0 visits its children. At a positive cube of side b_0, scan every triple, outputting exactly the genuine present zero triangles whose pair is still unsaturated; increment its counter immediately.

Every emitted witness is genuine, and leaves have disjoint tuple domains, so witnesses are distinct. Every positive cube emits at least once before returning. To see this, choose an active witness at entry. If no earlier descendant emits, its pair remains unsaturated and its path stays positive until a leaf emits it. If an earlier descendant emits, the assertion already holds. Depth-first execution ensures that no outside subtree changes the counters during this argument. A pair finally below T never saturated, so none of its witnesses could have been pruned; all were emitted. A pair at T has exactly T distinct stored witnesses. This proves the cap claim.

All queries have $b \ge b_0 \ge N^\alpha$. Since $n \ge 2$,

$$p^{d+3} \le p^3 n^c \le N^{c+3\log_2 p} \le b^{(c+3\log_2 p)/\alpha}.$$

Thus one fixed order exponent makes every detector call legal. There are no queries below this cutoff. Word sizes $\log b$ and $\log n$ differ by a fixed factor. At side b there are at most $(N/b)^3$ queries. Both their construction and detection cost at most $\widetilde{O}(N^3 b^{-\varepsilon_0})$, since $b^2 \le b^{3-\varepsilon_0}$. Summing over levels gives $\widetilde{O}(n^{3-\alpha\varepsilon_0})$, including negative queries and control.

There are $Z \le n^2 T = O(n^{3-\theta})$ outputs. Distinct positive leaves emit distinct tuples, so at most Z leaves are scanned. Their total cost is $\widetilde{O}(Z b_0^3) = \widetilde{O}(n^{3-\theta+3\alpha}) = \widetilde{O}(n^{3-\theta/2})$. This is dominated by the detector bound. Counters, lists, padding and coordinate operations fit the same bound; cubes can be streamed. If $b_0 = N$, the scan concerns a finite range depending on α, or follows the same estimates. No preprocessing universe is assumed. □

4 A charged quotient decomposition

Let M be composite, choose its least prime divisor p, and put $q = M/p$. Reduce A, B, C modulo p, writing a, b, c for labels in $Q = \mathbb{F}_p^d$. Choose a rational $0 < \theta < 1$, to be fixed in Section 6, and set

$$s = 2^{\lfloor \theta \lfloor \log_2 n \rfloor \rfloor}, \qquad L = \lfloor n/s \rfloor, \quad T = L + 1, \quad r = s^2. \tag{3}$$

Here $s = \Theta(n^\theta)$, $1 \le s \le n$, and $T = O(n^{1-\theta})$. Integer logarithms and rational arithmetic implement these parameters without floating-point rounding. Define $W_{ij} = \{k : a_{ik} + b_{kj} = c_{ij}\}$.

Apply Lemma 3.1. A pair with fewer than T stored witnesses is *light*; its list is complete and has size at most L. Put these triangles in R_{few}. All other pairs are *heavy*. For each heavy pair retain its genuine stored set S_{ij} of size $L + 1 > n/s$. The total number of stored incidences is $O(n^3/s)$.

Greedily hit these sets by middle vertices. If m unhit sets remain, their total incidences exceed mn/s, so some vertex hits more than m/s sets. Repeatedly choose a vertex of maximum current incidence degree. After $O(s\log(n+2))$ choices all sets are hit; denote the chosen set by H. Assign each heavy pair its first chosen hitter $h_{ij} \in S_{ij}$. In particular, this pivot is a real witness, so

$$c_{ij} = a_{ih_{ij}} + b_{h_{ij}j}. \tag{4}$$

For $s = 1$ no pair is heavy, since $T = n+1$. To implement the greedy rule, use incidence lists and a max heap of degrees, breaking ties by vertex index. Once a set is hit, delete its incidences and update degrees. Each incidence is deleted once; setup, updates and pivot assignment take $O((n^3/s + n^2 + n)\log(n+2))$ operations.

For each $h \in H$ and $k \in X$, sort the $|V|$ values $b_{hj} - b_{kj}$, keeping the list of indices for each value. Let F_{hk} contain the values of frequency strictly greater than n/r. There are at most r such values; give them a canonical lexicographic order. Write $t_{hk\ell}$ for the ℓ-th one, when defined, with $1 \le \ell \le r$. All tables take $\widetilde{O}(|H|n^2) = \widetilde{O}(sn^2)$ time and storage.

For each (h,i,k) look up $v = a_{ik} - a_{ih}$. If $v \notin F_{hk}$, scan its index list, of size at most n/r, and retain a triple (i,k,j) only if (i,j) is heavy and $h_{ij} = h$. Put these triples in R_{low}. Equation (4) and the matched value show they are quotient-zero. The assignment filter prevents duplicates from different pivots. The total scan and setup cost is

$$\widetilde{O}(|H|n^2 + |H|n^3/r) = \widetilde{O}(sn^2 + n^3/s). \tag{5}$$

For each $(h,\ell) \in H \times \{1,\dots,r\}$ form a masked graph $P_{h\ell}$ on the original vertex sets. Its present edges are exactly

$$\begin{aligned} &U_iV_j : (i,j) \text{ is heavy and } h_{ij} = h,\\ &U_iX_k : t_{hk\ell} \text{ is defined and } a_{ik} - a_{ih} = t_{hk\ell},\\ &X_kV_j : t_{hk\ell} \text{ is defined and } b_{hj} - b_{kj} = t_{hk\ell}. \end{aligned} \tag{6}$$

There are $|H|r = \widetilde{O}(s^3)$ pieces.

Lemma 4.1. *The quotient-zero triangles are the disjoint union of R_{few}, R_{low}, and the triangles of all $P_{h\ell}$. On a fixed piece there are potentials $\alpha_i, \beta_j, \tau_k \in Q$ such that every present edge satisfies*

$$a_{ik} = \alpha_i + \tau_k, \qquad b_{kj} = \beta_j - \tau_k, \qquad c_{ij} = \alpha_i + \beta_j. \tag{7}$$

Proof. A light pair contributes precisely its listed quotient-zero triangles. For a heavy quotient-zero triangle, its assigned $h = h_{ij}$ is unique, and (4) gives $a_{ik} - a_{ih} = b_{hj} - b_{kj}$. If this value is low frequency, the scan puts it in R_{low}. Otherwise it is in a unique slot ℓ of F_{hk}, and all three edges satisfy (6). Conversely, a triangle in a piece is quotient-zero by the same equations. Uniqueness of the assigned pivot and of the slot for fixed (h,k) proves disjointness.

For a fixed piece, take $\alpha_i = a_{ih}$, $\beta_j = b_{hj}$, and $\tau_k = t_{hk\ell}$ if defined, otherwise $\tau_k = 0$. The two incident-edge identities follow from (6); the outer identity follows from the assigned actual witness (4). If the slot is undefined, both incident edge kinds at X_k are absent, so its arbitrary potential makes no assertion about a present edge. □

This is the hitting/frequency mechanism of [1, Theorem 5.1], reconstructed over Q using only equality and subtraction. What changes in its construction is the capped acquisition of genuine witnesses, rather than cubic initial sorting. Materializing all pieces still costs $\widetilde{O}(s^3n^2)$; that charge is essential below.

5 Removing lift carries and encoding the kernel

Fix $P_{h\ell}$ and its potentials in Lemma 4.1. Lift $\alpha_i, \beta_j, \tau_k$ to vectors $u_i, v_j, z_k \in \mathbb{Z}_M^d$ by choosing coordinate representatives in $\{0, \dots, p-1\}$. These lifts need not define a homomorphism. On present edges put

$$A'_{ik} = A_{ik} - u_i - z_k, \qquad B'_{kj} = B_{kj} - v_j + z_k, \qquad C'_{ij} = C_{ij} - u_i - v_j. \tag{8}$$

Each vector reduces to zero modulo p by (7), so it belongs to $(p\mathbb{Z}_M)^d$. On any present triangle the lifted potentials cancel exactly:

$$A'_{ik} + B'_{kj} - C'_{ij} = A_{ik} + B_{kj} - C_{ij} \quad \text{in } \mathbb{Z}_M^d. \tag{9}$$

The additive map $p\mathbb{Z}_M \longrightarrow \mathbb{Z}_q$, $px \mapsto x \bmod q$, is well-defined and bijective: two representatives px, py agree modulo $M = pq$ exactly when $x = y \bmod q$. This works also when $p \mid q$. Consequently dividing all three adjusted vectors by p turns their zero-sum condition into the original condition (2). There is no assumption that the extension splits and no discarded carry.

Make a complete kernel instance over $\mathbb{Z}_q^{d+3}$. A present edge gets plus-sum label A'/p, B'/p, or $-C'/p$, with zero markers. An absent edge gets zero original coordinates and 1 in its own edge-kind marker. The marker argument in Section 2 proves that every zero triangle of this complete instance uses present edges. Together with (9), this gives an exact correspondence between its zero triangles and the original zero triangles contained in the piece. In particular the marker on the third edge is +1 as a plus-sum label; the sign convention for C does not alter the absence test.

Each query has the same vertex-side bound n as the original instance, and $q^{d+3} \le n^C$ for a fixed C depending on M, c, for $n \ge 2$. For example, the extra constant factor q^3 is at most $n^{3\log_2 q}$. Along recursion only three coordinates per layer are added, and the depth depends only on M. Thus the inductive order promise is valid in every call. We do not retain a large dimension while applying the kernel algorithm to tiny sides. All edge tests, potentials, adjusted coordinates, divisions, and markers take $\widetilde{O}(n^2)$ time per piece; this is an explicit complete graph, not an oracle representation.

6 The algorithm and its total cost

We prove Theorem 1.1 by induction on M. Empty parts return NO. If the input group is trivial, a nonempty complete graph returns YES. For $n \le 1$, read and check its at most one triangle directly. Apply the coordinate embedding of Section 2 otherwise. For prime M, use (1), setting $\delta_M = \varepsilon_0$.

For composite M, let p be its least prime divisor, $q = M/p \ge 2$, and use the already established algorithm at exponent q. Set

$$\theta = \delta_q/12, \qquad \alpha = \theta/6, \qquad \delta_M = \varepsilon_0\delta_q/144 = \delta_q/144000. \tag{10}$$

Construct the capped lists, exception sets and pieces of Section 4. Check every exception in the original group, returning YES if it satisfies (2). For each piece construct the complete kernel instance of Section 5 and run the exponent-q algorithm. Return YES if any call does, and NO otherwise.

Correctness follows from the disjoint partition in Lemma 4.1 and the kernel correspondence. Every original zero triangle is quotient-zero, so it is checked directly or represented in a kernel query. Conversely,

exception checks are made in the original group, and a positive kernel query has no absent edge and satisfies (9). All recursive decisions are correct by induction. Only decision is needed from the kernel algorithm.

For clarity the full ledger, including every complete graph construction, is

Operation	Charged time, up to fixed polylogarithms
Capped quotient witnesses	$n^{3-\varepsilon_0\theta/6}$
Lists, hitting set, light exceptions	$n^3/s + n^2$
Frequency tables, low exceptions	$sn^2 + n^3/s$
All materialized kernel graphs	s^3n^2
All recursive kernel calls	$s^3n^{3-\delta_q}$

The $\log(n+2)$ factors in $|H|$ and sorting are included in the notation. In particular we multiply the entire inductive cost by the number of pieces; there is no amortized free call or uncharged initialization. Input reading, coordinate embedding, direct exception tests, control, and termination are covered by the displayed terms. Streaming one complete piece at a time is sufficient. Stored witness lists, frequency tables, counters and incidence structures have polynomial size, so their indices occupy a constant number of logarithmic words. No unit-cost long integer operations are used.

Because $s = \Theta(n^\theta)$, the possible exponent savings in this ledger are at least

$$\varepsilon_0\theta/6, \quad \theta, \quad 1-\theta, \quad 1-3\theta, \quad \delta_q - 3\theta = 3\delta_q/4.$$

Here $0 < \delta_q \le \varepsilon_0 \le 1$, so the smallest is $\varepsilon_0\delta_q/72$. Taking half of it in (10) leaves slack to absorb every fixed polylogarithm, even when its exponent or constant depends on M, c. All rounding in (3) changes only fixed multiplicative constants. Fixed-prime exceptional ranges of the backend, and any finite small-n cases, are absorbed by direct enumeration with constants depending on the fixed parameters. This proves the stated time bound. Iterating (10) until a prime remains gives $\delta_M = 10^{-3}(144000)^{1-\Omega(M)}$, independently of c.

Uniformity here means an explicit deterministic procedure for the compact input: the dimension is read, the least prime divisor and rational parameters are computable from M, and every loop and recursive call is specified. There is no advice depending on n, no preprocessed universe, and no supplied order exponent required by the algorithm. All recursive exponent parameters are smaller, so termination follows as well. This completes the proof.

7 Verification and limitations

The theorem is a decision algorithm on complete compact inputs for fixed M, with an explicit saving depending on M. Its proof uses the stated elementary-group theorem of [2] and the existing matrix primitive. The source's full chain of matrix dependencies is used as prior work; a formal verification of that chain is outside this paper.

Exact Python checks supplement the argument. A separately written verifier checked 512 capped-listing instances on sides in $\{0,1,2,3\}$ with varying caps and traversal orders; 676,333 one-coordinate potential/kernel cases for composite $M \le 36$ and proper prime divisors; and 3,548,440 marker cases for $2 \le q \le 36$. It also checked 360 seeded graph decompositions with sides 1–6 and dimensions 1–3, yielding 6,850 explicit kernel instances, and 2,047 exact rational exponent ledgers for $2 \le M \le 2048$. The graph tests include ordinary and dense quotient-gauge inputs. Their detectors are brute force; these are finite

semantic checks, neither a benchmark of the fast algorithm nor a proof of the infinite theorem. The supplied verification scripts and their coverage records are suitable for reproducing these finite checks locally.

AI methodology. OpenAI Codex was used substantially throughout the research: to develop and test the quotient/kernel algorithm, reconstruct supporting prior machinery, search and compare primary literature, write exact verification programs, and draft this paper. A separate model context performed an internal mathematical and novelty review and wrote the additional verifier. This assistance materially affected the arguments, experiments, literature comparisons, and exposition throughout the paper. The sole listed human author is responsible for checking the content, dependencies, references, and originality before dissemination.

References

[1] Timothy M. Chan, Virginia Vassilevska Williams, and Yinzhan Xu. 2023. Fredman's Trick Meets Dominance Product: Fine-Grained Complexity of Unweighted APSP, 3SUM Counting, and More. arXiv:2303.14572 [cs.DS] doi:10.48550/arXiv.2303.14572 Version 1, 25 March 2023.

[2] Ruhollah Majdoddin. 2026. Thin matrix products up to $D \leq N^{0.172}$ and lossless reductions for 3SUM, Exact Triangle and APSP, with 3SUM over abelian groups and a dichotomy for coordinatewise problems. arXiv:2610.17420 [cs.DS] doi:10.48550/arXiv.2610.17420 Version 1, 12 October 2026.

[3] Nathan Sheffield, Virginia Vassilevska Williams, and Zoe Xi. 2026. The Limits of Black-Box Reductions for All-Pairs Triangle Detection. arXiv:2608.19092 [cs.DS] doi:10.48550/arXiv.2608.19092 Version 1, 19 August 2026.